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Studierichting: Informatica
Oefeningenreeks 6B nr. 17.6

$1^2 \cdot 4 + 2^2 \cdot 5 + 3^2 \cdot 6 + \dots$

$$
\begin{aligned}
s_n &= \sum_{i=1}^{n} i^2(i+3) \\
&= \sum_{i=1}^{n} i^3 + 3\sum_{i=1}^{n} i^2 \\
&= \left(\frac{n(n+1)}{2}\right)^2 + 3 \cdot \frac{n(n+1)(2n+1)}{6} \\
&= \frac{n^2(n+1)^2}{4} + \frac{n(n+1)(2n+1)}{2} \\
&= \frac{n(n+1)}{2}\left(\frac{n(n+1)}{2} + (2n+1)\right) \\
&= \frac{n(n+1)}{2}\left(\frac{n(n+1) + 2(2n+1)}{2}\right) \\
&= \frac{n(n+1)}{2}\left(\frac{n^2 + n + 4n + 2}{2}\right) \\
&= \frac{n(n+1)}{2}\left(\frac{n^2 + 5n + 2}{2}\right) \\
&= \frac{n(n+1)(n^2 + 5n + 2)}{4}
\end{aligned}
$$

$n(n+1)$ is deelbaar door 2: Triviaal

$(n^2 + 5n + 2)$ is deelbaar door 2:

$$
\begin{cases}
n \text{ even: } n^2 \text{ en } 5n \text{ even} \Rightarrow n^2 + 5n + 2 \text{ even} \\
n \text{ oneven: } n^2 + 5n \text{ oneven} + \text{ oneven} = \text{ even}
\end{cases}
$$

$\Rightarrow$ deelbaar door 2

References